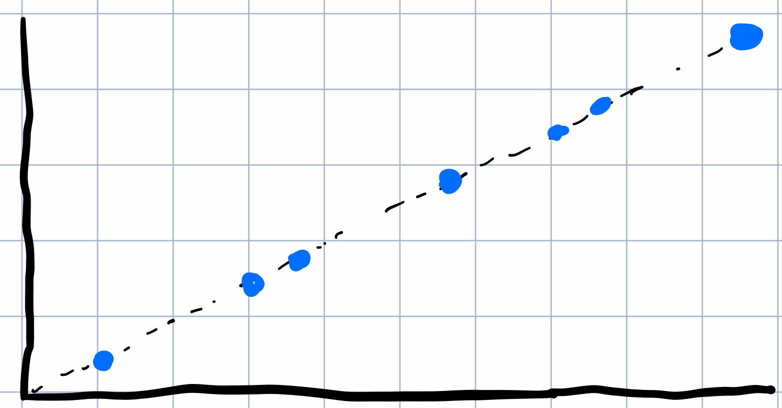
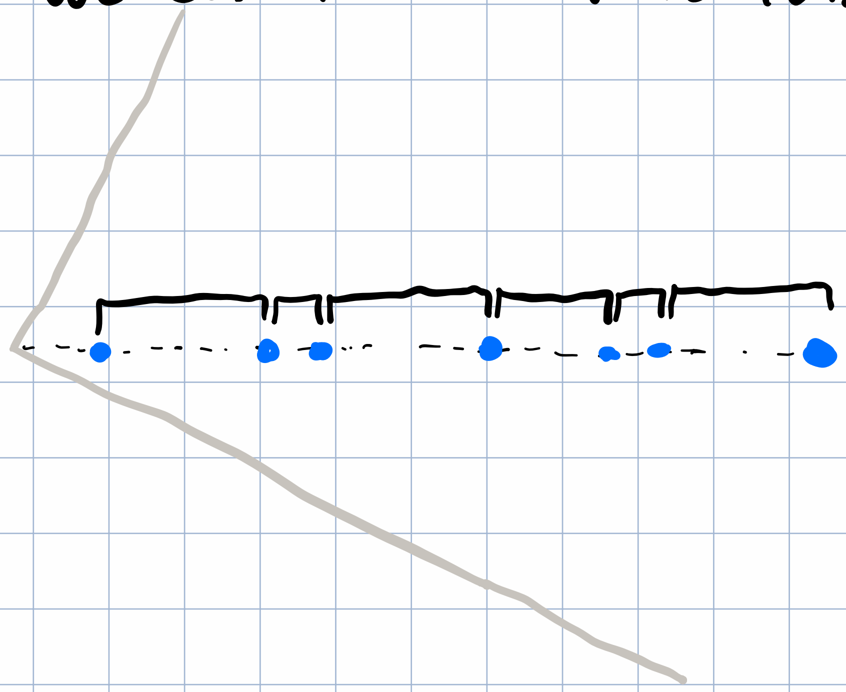
1) If all of our data lies exactly on a 2-D line, there is only one principal component has a non-zero component because when we orient to the line that the data is on the only variation is how spread out the data points are. For instance, in the graph below



we can reorient to that line.

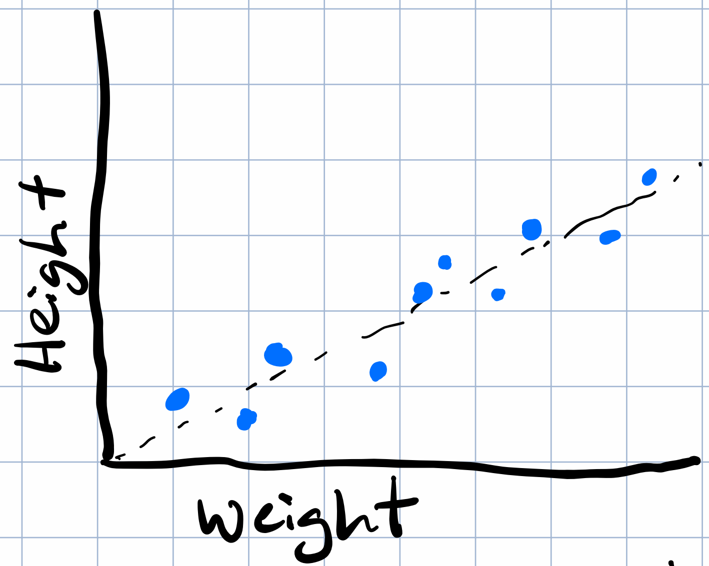


When we do the data is purely one-dimensional.

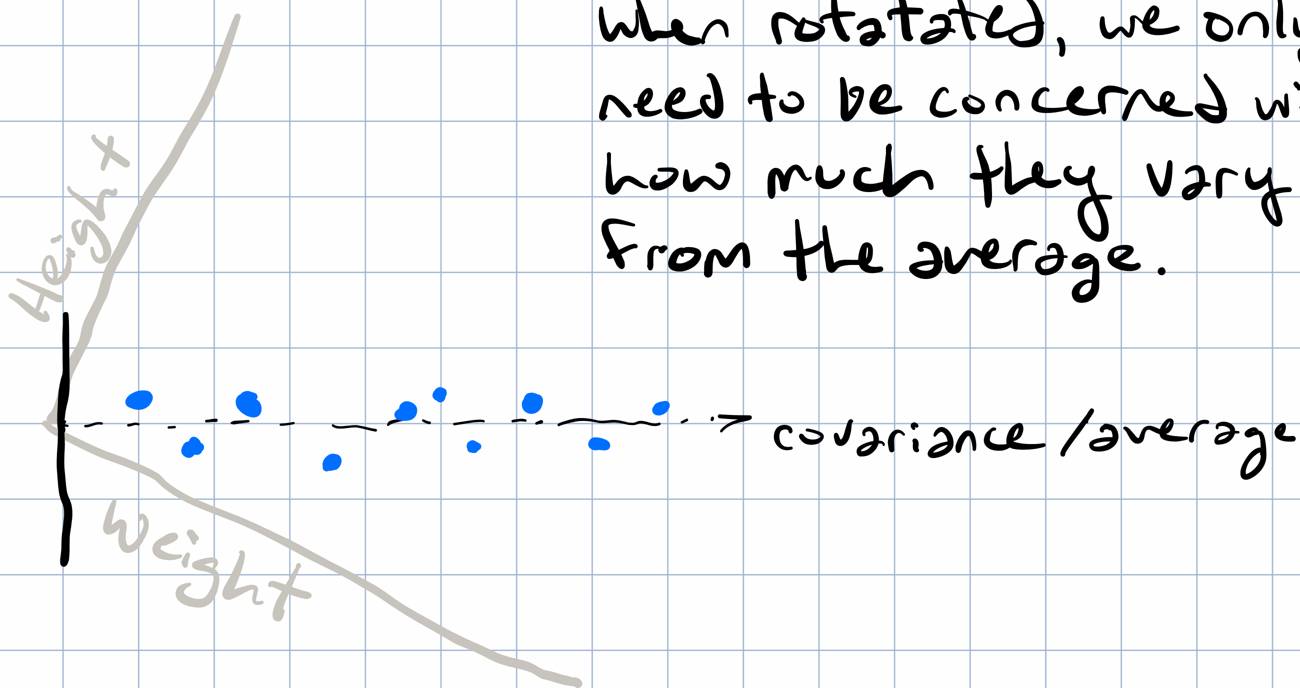
If the data were just a little bit off of that line, we can use that line as a good approximation.

2) Suppose Height and Weight have a large positive covariance. What does that mean geometrically about the data cloud?

The data cloud would look almost like a line in the positive direction. It might look something like this



When rotatated, we only need to be concerned with how much they vary from the average.



3) The right singular vectors of the centered data matrix become the principal components because we are reorienting to the covariance by rotating our coordinants to align with the right singular vectors. The principal components define our new coordinate system